ADS401 EVALUATION METHODS AND STATISTICS

Usability Evaluation of the XJTLU Museum AI System

Task success, usability, and visitor barriers in a short museum session

PRESENTATION 2

Research design and pilot plan

GROUP H

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3-minute presentation

A public AI guide must work without a manual

The practical problem

Visitors may not know how to start the guide, what to ask, where to go next, or how to recover from an unclear response.

The Museum needs direct evidence about the actions visitors actually perform.

RQ1 · Task success

Can visitors complete the four core Museum AI tasks?

RQ2 · Usability

How usable and efficient is the system according to task performance and SUS?

RQ3 · Barriers

Which onboarding, navigation, answer, or recovery barriers recur?

TARGET USERS XJTLU students and staff, 18+, with mixed digital confidence and museum experience

Four tasks follow the visitor journey

TASK 01

Start

Open the guide and select an exhibit.

TASK 02

Ask

Ask one exhibit question and identify the answer.

TASK 03

Navigate

Move to the next recommended stop.

TASK 04

Recover

Resolve an unclear or unsuitable answer.

METHOD

Moderated task-based usability test with think-aloud comments.

SAMPLE

12-15 XJTLU students or staff; one 15-minute session.

RATIONALE

Observed behaviour shows where users hesitate, fail, or need help.

One session, five steps, one aligned questionnaire



Participant questionnaire flow

A

Background

Role and prior Museum AI use

B

Task experience

Completion, ease, and confidence

C

SUS

10 standard usability items

D

System feedback

Content, navigation, recovery, trust, open

10 bilingual questions are collected in Wenjuanxing; the dashboard generates frequencies, charts, and downloadable data.

Survey: v.wjx.cn/vm/QZpCC3h.aspx · No names, student numbers, passwords, photographs, or personal museum records are collected.

Observed behaviour and self-report answer different questions

Observed

- Task success: independent, assisted, failed
- Time on task in seconds
- Errors and moderator help

Self-report

- SUS score
- Perceived ease after each task
- Confidence and trust

Qualitative

- Think-aloud hesitation
- Repeated workarounds
- Open comments and barriers

Analysis

Binomial test · Wilcoxon SUS test · Spearman correlations · coded themes

H1 ≥80% independent task success · H2 SUS ≥68 · H3 ease and confidence relate to SUS · H4 recurring barriers

The study package is ready for a small pilot

READY

Participant materials

- Online questionnaire
- Information sheet
- Consent form
- Moderator script

CHECK

System access

- Confirm current AI version
- Prepare test account if needed
- Prepare fallback walkthrough

NEXT

Pilot and main sessions

- Pilot with two participants
- Revise unclear task wording
- Run 12-15 sessions

DATA STATUS

The survey is live. Pilot and participant sessions are pending, so we present the design, not invented findings.

The output is a concrete usability diagnosis

01 Task success profile for start, ask, navigate, and recovery.

02 Evidence about where time, errors, or moderator help increase.

03 SUS, ease, and trust scores from the aligned questionnaire.

04 Prioritised recommendations for the Museum AI interface.

Q&A

Thank you. We welcome questions about task design, sampling, measures, and the pilot plan.

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